

PROJECT DESIGN PHASE-II

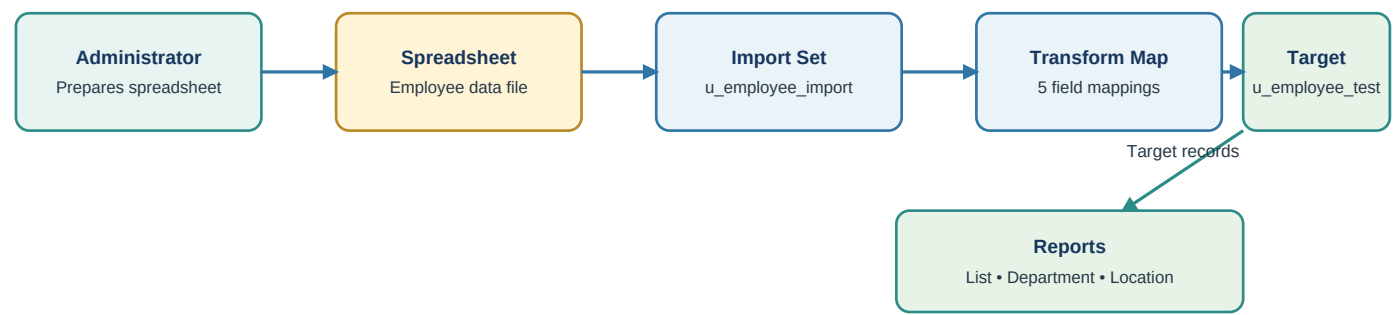
Technology Stack (Architecture & Stack)

Date	03 October 2026	Team ID	SWTID-2026-7959
Project Name	Import Data Using Transform Maps (Spreadsheet)	Maximum Marks	4 Marks

Technical Architecture

The architecture below represents the documented ServiceNow import workflow. Details marked **Proposed** are recommendations for documenting or strengthening the design, not claims that those technologies are currently configured.

Proposed Technical Architecture — ServiceNow Spreadsheet Import



Environment: ServiceNow instance (hosted environment); external APIs and ML models are not specified for this project.

Infrastructure demarcation: Spreadsheet preparation occurs on the administrator's local system. Import Set, Transform Map, target table, and reports operate within the ServiceNow instance. Specific hosting configuration is not confirmed in the supplied project details.

TABLE 1 — COMPONENTS & TECHNOLOGIES

Known project components are distinguished from proposed technologies and unspecified integrations.

S.No.	Component	Description	Technology
1	User Interface	Administrator uses ServiceNow forms/lists and spreadsheet software to prepare and review data.	ServiceNow UI; spreadsheet application (specific product not specified)
2	Application Logic — Import	Loads spreadsheet rows into the staging/import-set table.	ServiceNow Import Set functionality
3	Application Logic — Transform	Maps source columns to target fields and transfers values.	ServiceNow Transform Map
4	Application Logic — Coalesce	Matches incoming rows to existing records to reduce duplicate creation.	ServiceNow coalesce setting; screenshot shows Employee Name enabled
5	Staging Database/Table	Temporarily holds imported rows before transformation.	Employee Import — u_employee_import
6	Target Database/Table	Stores transformed employee records.	Employee Test — u_employee_test
7	Field Mapping	Five source fields are mapped to target fields.	Employee ID, Name, Email, Department, Location
8	Reporting	Provides a list and grouped summaries of imported employee data.	Employees List; Employees by Department; Employees by Location
9	File Storage	Spreadsheet is selected/uploaded by the administrator; persistent file storage configuration is not specified.	Local spreadsheet before import; ServiceNow attachment/storage details not confirmed
10	External API-1	No external API is documented for the current workflow.	Not specified / not currently evidenced
11	External API-2	No second external API is documented for the current workflow.	Not specified / not currently evidenced
12	Machine Learning Model	No ML model is required by the documented import and reporting workflow.	Not applicable to current project scope
13	Infrastructure	The application is accessed through a ServiceNow instance.	ServiceNow hosted instance; exact cloud/region/server configuration not specified

TABLE 2 — APPLICATION CHARACTERISTICS
 Security, scalability, availability, and performance characteristics for the documented workflow

S.No.	Characteristic	Description	Technology / Implementation
1	Open-Source Frameworks	No separate open-source framework is documented. The workflow uses native ServiceNow import, transform, table, and reporting features.	No additional framework specified
2	Security Implementations	Recommended: restrict access to authorized ServiceNow roles, apply least privilege, avoid exposing employee data, and follow instance security policies. Exact controls are not confirmed.	Proposed: role-based access control; review ACLs and audit settings
3	Scalable Architecture	The workflow separates staging, transformation, target storage, and reporting, which makes each step easier to maintain. Large-volume limits should be tested.	ServiceNow Import Sets + Transform Maps; volume testing proposed
4	Availability	Availability depends on the ServiceNow instance and user access. Redundancy, load balancing, and service-level configuration are not specified.	ServiceNow platform; deployment details not specified
5	Performance	The known sample contains six records. For larger spreadsheets, validate import duration, transform time, and report response time before wider use.	Six sample records SB0001–SB0006; performance benchmark not specified
6	Data Integrity	Compare Employee ID, Name, Email, Department, and Location in the target table against the source spreadsheet after transformation.	Five mapped fields; validation checklist recommended
7	Maintainability	Keep field mapping names, coalesce behavior, and report definitions documented; retest after configuration changes.	Transform Map configuration and report definitions
8	Monitoring / Auditability	Review import and transform outcomes, identify failed or mismatched records, and rerun after correcting source data or configuration.	ServiceNow import/transform result review; exact logging setup not specified

References / Project evidence: Project screenshots supplied in the conversation showing the Transform Map, field mappings, Employee Import table, Employee Test table, six sample employee records, and the three report definitions. Template reference links: <https://c4model.com/> ; <https://www.ibm.com/cloud/architecture> ; <https://aws.amazon.com/architecture> . These architecture references are background guidance; they do not imply that IBM or AWS services are used in this project.